

Data Source & Attribution

This project uses the **HDHI Cardiac Dataset** available on Kaggle.

<https://www.kaggle.com/datasets/ashishsahani/hospital-admissions-data>

If used in academic research, please cite:

Bollepalli, S.C. et al.

An Optimized Machine Learning Model Accurately Predicts In-Hospital Outcomes at Admission to a Cardiac Unit.

Diagnostics 2022, 12, 241.

<https://doi.org/10.3390/diagnostics12020241>

Note:

- This dataset is intended for research and educational purposes
- Commercial use requires explicit permission from the data providers
- This project focuses on data analysis and does not claim ownership of the dataset.

Section 0: Data Preparation (ETL)

Overview

This project integrates three healthcare-related datasets to build a unified analytical environment for cardiac outcome analysis:

- Admissions Dataset** → hospitalizations, diagnoses, lab values, and outcomes
- Mortality Dataset** → patients brought dead to the hospital
- Pollution Dataset** → daily air quality and environmental measurements

ETL Process

The data pipeline follows a staging-to-production workflow:

Raw CSV Files → Staging Tables → Data Cleaning → Production Tables

Key ETL operations included

- Bulk loading CSV files into staging tables
- Data type conversion using `TRY\_CAST()`
- Missing/invalid value handling
- Standardization of categorical fields (gender, rural/urban)
- Range validation for clinical and environmental variables

Final Tables

Table	Purpose
`admissions`	Cardiac hospitalization records
`mortality`	Brought-dead mortality records
`pollution`	Daily air quality and weather data

Why This Matters

A clean and structured data foundation is essential for reliable downstream analysis, including:

- mortality analysis
- comorbidity risk assessment
- clinical outcome evaluation
- environmental impact analysis

Section 1: Data Exploration & Quality Assessment

Overview

Before starting analytical modeling, an initial data audit was performed to assess dataset completeness, structure, and overall quality.

Key Validation Checks

- Verified dataset size across all three tables
- Checked missing values in critical clinical variables
- Validated categorical field consistency
- Generated statistical summaries for core clinical indicators
- Confirmed the temporal coverage of admission records

Key Findings

- 15,757** cardiac admission records available for analysis
- 359** additional mortality records (brought-dead cases)
- 2,208** daily environmental observations for pollution analysis

Patient Profile Summary:

- Average patient age: **61 years**
- Age range: **4–110 years**

Clinical Severity Indicators:

- Average ejection fraction (EF): **43.5%**
- EF range: **14–60%**

Hospitalization Metrics:

- Average length of stay: **6 days**
- Average ICU stay: **3 days**
- Maximum hospital stay: **98 days**

Study Period:

- Admission records span from **January 2017 to December 2019**

Why This Matters: This validation step confirmed sufficient data volume, acceptable clinical coverage, and multi-year temporal consistency—providing a reliable foundation for downstream mortality, comorbidity, and environmental impact analyses.

Section 2: Patient Demographics

Overview

This section analyzes the demographic composition of cardiac admissions to understand population distribution and identify demographic disparities in outcomes.

Key Analysis

- Evaluated gender distribution of admitted patients
- Compared urban and rural patient populations
- Segmented patients into clinical age groups
- Assessed mortality differences by geographic background

Key Findings

Gender Distribution:

- **63.4%** of patients were male (**9,990** cases)
- **36.6%** were female (**5,767** cases)

Geographic Distribution:

- **76.6%** of admissions came from urban areas (**12,077** patients)
- **23.4%** came from rural areas (**3,680** patients)

Age Distribution:

- The highest admission burden was in the **60–69 age group** (**4,957** patients)
- Cardiac admissions increased significantly after age 50

Clinical Severity by Age:

- Average ejection fraction declined from **49.4%** (<40 years) to **42.2%** (80+ years)
- Average hospital stay increased from **5 days** to **7 days** with age

Mortality by Location:

- Rural mortality rate: **7.55%**
- Urban mortality rate: **6.85%**

Why This Matters

Demographic analysis helps identify high-risk patient groups and healthcare disparities. The findings highlight higher cardiac burden in older adults, stronger male predominance, and increased mortality among rural patients, which may reflect delayed access to specialized cardiac care.

Section 3: Admission Patterns & Length of Stay

Overview

This section analyzes hospital admission patterns and length of stay (LOS) to understand patient flow, care intensity, and outcome-related differences in hospitalization.

Key Analysis

- Compared Emergency vs OPD admission volumes and LOS
- Evaluated monthly admission trends and mortality patterns
- Analyzed LOS and ICU usage by patient outcome
- Assessed LOS differences by gender and admission type

Key Findings

Admission Type Distribution:

- **69.3%** of admissions were Emergency (**10,924 cases**)
- **30.7%** were OPD (**4,833 cases**)
- Emergency cases had longer stays (**7 days vs 5 days**)

Monthly Admission Trends:

- Admissions remained relatively stable (~500–870 per month)
- Peak admissions observed in **Jan 2019 (870 cases)**
- Mortality counts fluctuated across months without extreme spikes
- Average LOS remained consistent (~6 days), indicating stable care patterns

Length of Stay by Outcome:

- **Discharged patients:** avg LOS **6 days**, ICU **3 days**
- **Expired patients:** shorter LOS (**5 days**) but higher ICU usage (**4 days**)
- **DAMA patients:** similar LOS to expired patients (~5 days)

LOS by Gender & Admission Type:

- Emergency female patients had the longest stays (**7 days**)
- Emergency male patients averaged **6 days**
- OPD admissions had shorter stays (**~5 days**) regardless of gender

Why This Matters

Understanding admission patterns helps identify hospital workload dynamics and care intensity. The dominance of emergency admissions and their longer stays indicate higher acuity cases, while shorter LOS in expired patients suggests rapid clinical deterioration. These patterns are critical for hospital resource planning and risk stratification.

Section 4: Outcomes & Mortality Analysis

Overview

This section evaluates patient outcomes and mortality patterns to identify key risk factors associated with in-hospital death.

Key Analysis

- Assessed overall outcome distribution (Discharge, Expiry, DAMA)
- Compared mortality rates by gender
- Analyzed mortality trends across age groups
- Evaluated mortality differences by admission type
- Examined monthly mortality trends

Key Findings

Outcome Distribution:

- **87.3%** of patients were discharged (**13,756 cases**)
- **7.01%** resulted in in-hospital death (**1,105 cases**)
- **5.69%** left against medical advice (DAMA)

Mortality by Gender:

- Male mortality rate: **7.03%**
- Female mortality rate: **6.99%**
- Mortality rates were nearly identical across genders

Mortality by Age:

- Mortality increased significantly with age:
 - **<50 years:** 5.13%
 - **60–69 years:** 6.68%
 - **70–79 years:** 9.23%
 - **80+ years:** **11.68%** (highest risk group)

Mortality by Admission Type:

- **Emergency admissions:** 9.86% mortality
- **OPD admissions:** 0.58% mortality
- Emergency cases show substantially higher risk, reflecting higher clinical severity

Monthly Mortality Trends:

- Mortality rates fluctuated between **~3.6% and 12.6%**
- Highest observed in **April 2017 (12.65%)** and **January 2018 (10.87%)**
- No consistent upward or downward trend, suggesting episodic variation

Why This Matters

Mortality analysis highlights critical risk factors in cardiac care. Age and emergency admission status emerged as the strongest predictors of death, while gender showed minimal impact. These findings are essential for clinical prioritization, early intervention strategies, and hospital resource allocation.

Section 5: Comorbidity Analysis (Risk Factors)

Overview

This section evaluates the prevalence of major comorbidities and their impact on patient outcomes, particularly in-hospital mortality.

Key Analysis:

- Measured prevalence of key chronic conditions (DM, HTN, CAD, CKD, etc.)
- Calculated comorbidity burden per patient (comorbidity score)
- Compared mortality rates between patients with and without specific conditions

Key Findings

Comorbidity Prevalence:

- **Coronary Artery Disease (CAD): 67%** (most common)
- **Hypertension: 48.6%**
- **Diabetes: 32.3%**
- **Prior Cardiomyopathy: 15.4%**
- **CKD: 9.8%**
- Smoking and alcohol prevalence were relatively low (**5–6.5%**)

Comorbidity Burden:

- Most patients had **1–3 comorbidities**
- Mortality did not increase linearly with comorbidity count
- Highest observed mortality at score **1 (9.54%)**, then fluctuates across higher scores

Impact on Mortality:

- **CKD** showed the strongest association with mortality:
 - With CKD: **14.39%**
 - Without CKD: **6.21%**

- **Diabetes and Hypertension** showed slightly lower mortality in affected patients (likely due to treatment/monitoring bias)
- **Smoking** group showed lower mortality (**2.77% vs 7.24%**), suggesting possible data imbalance or underreporting

Why This Matters

Comorbidity analysis identifies high-risk patient groups. CKD emerges as a critical predictor of mortality, while other risk factors require careful interpretation due to potential data bias. These insights are essential for risk stratification and targeted clinical intervention.

Section 6: Cardiac Diagnosis Analysis

Overview

This section analyzes major cardiac diagnoses, their clinical severity, and associated outcomes using ejection fraction (EF) and length of stay as key indicators.

Key Analysis

- Attempted prevalence and mortality analysis for major diagnoses
- Grouped patients by primary cardiac condition
- Compared EF and hospital stay across diagnosis categories

Key Findings

Data Limitation:

- Prevalence and mortality outputs returned **NULL values**
- This indicates a **data quality or transformation issue** (likely during ETL or casting of diagnosis columns)
- As a result, diagnosis frequency and mortality comparisons could not be reliably computed

Clinical Severity (EF-Based):

- **Lowest EF observed in heart failure groups:**
 - HFrEF: **36%**
 - HFnEF: **35.5%**
- **STEMI patients** also showed reduced EF (**39%**)
- **Higher EF in less severe conditions:**
 - ACS: **46.2%**
 - Atrial Fibrillation: **48.5%**
 - Other diagnoses: **49.2%**

Length of Stay:

- Heart failure groups had the longest stays (**~7 days**)
- ACS patients had shorter stays (**~5 days**)
- Most other groups averaged **6–7 days**

Why This Matters

Despite incomplete diagnosis prevalence data, EF-based analysis confirms expected clinical patterns: heart failure and STEMI patients present with more severe cardiac dysfunction and longer hospital stays. Fixing the underlying data issue would enable full diagnostic and mortality comparisons.

Section 7: Laboratory Values Analysis

Overview

This section evaluates key laboratory biomarkers to understand their relationship with disease severity, hospital stay, and mortality outcomes.

Key Analysis

- Compared lab values across patient outcomes
- Assessed impact of anaemia severity
- Evaluated AKI as a mortality risk factor
- Analyzed effect of elevated cardiac enzymes

Key Findings

Outcome-Based Clinical Differences:

- Non-survivors (**EXPIRY**) showed significantly worse lab profiles:
 - Higher BNP: **1379** vs **735** (discharged)
 - Lower EF: **35%** vs **44%**
 - Higher creatinine: **2.07** vs **1.27**
- Indicates strong association between **organ dysfunction and mortality**

Anaemia Impact:

- Anaemia associated with worse outcomes:
 - Mild/Moderate: **11.4% mortality**
 - Severe: **9.5% mortality**
 - No anaemia: **6.1% mortality**
- Anaemic patients also had **longer hospital stays**

AKI as a Major Risk Factor:

- AKI patients had **very high mortality: 16.2% vs 4.4%**
- Markedly worse kidney markers (creatinine, urea)
- Longer hospital stay (**8 vs 5 days**)

Cardiac Enzymes:

- Patients with elevated enzymes had:
 - Higher mortality (**10.9% vs 6.0%**)
 - Lower EF and higher BNP
- Confirms role of **myocardial injury in poor outcomes**

Why This Matters

Laboratory markers—especially BNP, creatinine, and cardiac enzymes—are strong predictors of mortality. AKI and anaemia emerge as critical risk factors, reinforcing their importance in early risk stratification and clinical decision-making.

Section 8: Pollution–Admission Correlation

Overview

This section explores the relationship between environmental factors—especially air pollution—and cardiac hospital admissions, providing a unique population-level perspective.

Key Analysis

- Linked daily admission data with pollution metrics
- Grouped admissions by AQI severity levels
- Identified extreme pollution events
- Evaluated temperature impact on admission volume

Key Findings

Air Quality and Admissions:

- Clear upward trend between pollution severity and admissions:
 - **Good AQI:** ~27 admissions/day
 - **Unhealthy:** ~40 admissions/day
 - **Hazardous:** ~52 admissions/day
- Indicates strong **positive correlation between AQI and cardiac admissions**

Extreme Pollution Events:

- Some days showed extremely high AQI (>700) with concurrent admissions and deaths
- Example:
 - **AQI 797 → 18 admissions, 1 death**
- High PM2.5 and PM10 levels were dominant pollutants

Temperature Effects:

- Admissions increased with temperature:
 - Cold days: **~24 admissions/day**
 - Hot days (>38°C): **~46 admissions/day**
- Suggests **heat stress as an additional cardiac trigger**

Pollution Category Impact:

- “Very Unhealthy” and “Hazardous” days had the highest burden
- Nearly **2x increase in admissions** compared to clean air days

Why This Matters

Environmental exposure—especially air pollution and extreme heat—plays a measurable role in cardiac events. This highlights the importance of integrating environmental data into public health planning, early warning systems, and hospital resource allocation.

Section 9: Advanced Analytics (Window Functions & Subqueries)

Overview

This section applies advanced SQL techniques to uncover temporal trends, identify high-risk patients, and perform clinical risk stratification using window functions and analytical queries.

Key Analytical Focus

- Identified peak admission periods using ranking functions
- Analyzed cumulative and smoothed admission trends over time
- Detected extreme clinical outliers (BNP)
- Stratified patients based on cardiac function (EF)
- Investigated patterns in DAMA (Discharged Against Medical Advice) cases

Key Findings

Admission Trends & Seasonality:

- Peak admission months:
 - **January 2019 (870 admissions)** – highest volume
 - Followed by **February 2019** and **January 2018**
- Clear **seasonal pattern**, with higher admissions in winter months

Longitudinal Admission Growth:

- Total admissions increased steadily to **15,000+ cases** over the study period
- No major disruptions observed → indicates **consistent hospital demand**

Smoothed Admission Trends (7-Day Average):

- Daily admissions typically ranged:
 - **~10–15 (low periods)**
 - **~20–30 (peak periods)**
- Rolling average confirms:
 - Reduced daily noise
 - More stable underlying trend
 - Alignment with seasonal patterns

High-Risk Clinical Outliers (BNP Analysis):

- Extreme BNP values observed at **5000**
- Deviation from mean: **+4182**
- Associated with:
 - Low EF values
 - Higher proportion of **EXPIRY outcomes**

Ejection Fraction (EF) Risk Stratification:

- **Severely Reduced (<30%)**
 - Mortality: **16.08%**
 - Highest BNP and longest stays
- **Reduced (30–39%)**
 - Mortality: **9.31%**
- **Mildly Reduced (40–49%)**
 - Mortality: **4.97%**
- **Preserved (≥50%)**
 - Mortality: **3.38%** (lowest risk group)

DAMA (Discharged Against Medical Advice) Patterns:

- Highest frequency group:
 - **Male, Urban, Emergency admissions (372 cases)**
- General trends:
 - Emergency > OPD
 - Urban > Rural
 - Slight male predominance
- Average patient profile:
 - Age: **~60–64 years**
 - LOS before leaving: **4–7 days**

Why This Matters

This section demonstrates how advanced analytical techniques can transform raw hospital data into actionable insights:

- Identifying **seasonal admission surges** supports resource planning
- Rolling averages provide **stable operational metrics** for monitoring trends
- BNP and EF analysis highlight **strong clinical predictors of mortality**
- DAMA insights reveal a **hidden high-risk patient group** requiring targeted intervention

Together, these analyses bridge clinical understanding with data science—enabling both **predictive modeling** and **healthcare decision support systems**.

Section 10: Mortality Analysis (Brought-Dead Cases)

Overview

This section analyzes patients who were **brought dead (DOA)** to the hospital and integrates these records with in-hospital mortality to provide a complete picture of cardiac death burden.

Key Analytical Focus

- Quantified brought-dead cases
- Examined demographic characteristics (age, gender)
- Analyzed monthly mortality trends
- Combined DOA and in-hospital deaths for total mortality burden

Key Findings

Brought-Dead Overview:

- Total brought-dead cases: **359 patients**

Demographic Distribution:

- **Male:** 236 cases (avg age: **61**)
- **Female:** 123 cases (avg age: **65**)
- Age range:
 - Youngest: **5 years**
 - Oldest: **95 years**

Temporal Trends:

- Highest monthly DOA cases:
 - **December 2018 (33 cases)**
 - **December 2017 (28 cases)**
- Noticeable fluctuations across months, with some peaks in:
 - Winter periods
 - Late-year months

Total Cardiac Mortality Burden:

- **In-hospital deaths: 1,105**
- **Brought-dead (DOA): 359**
- **Total cardiac deaths: 1,464**

Why This Matters

Including brought-dead cases provides a **more complete and realistic estimate of cardiac mortality**, which is often underestimated when only hospital outcomes are analyzed.

- Highlights **pre-hospital mortality burden**
- Suggests potential delays in **access to care or emergency response**
- Enables better alignment with **public health and emergency system improvements**

This integration strengthens the overall analysis by bridging **clinical outcomes inside the hospital** with **real-world mortality outside hospital settings**.

conclusion

This project provides a comprehensive, data-driven analysis of cardiac hospitalizations by integrating clinical, demographic, and environmental datasets into a unified analytical framework .

The findings highlight several critical insights:

- **Age and emergency admission** emerged as the strongest predictors of mortality
- **Chronic Kidney Disease (CKD)** and **Acute Kidney Injury (AKI)** showed the highest association with in-hospital death
- **Laboratory markers** such as BNP, creatinine, and cardiac enzymes demonstrated strong correlation with disease severity
- **Ejection Fraction (EF)** proved to be a powerful clinical stratification tool, with mortality increasing significantly as EF decreases
- **Environmental factors**, particularly **air pollution (AQI)** and **high temperature**, showed a clear relationship with increased cardiac admissions

In addition, incorporating **brought-dead (DOA)** cases revealed that a substantial portion of cardiac mortality occurs **before hospital admission**, emphasizing gaps in early intervention and emergency response systems.

From a data perspective, this project demonstrates the value of:

- Structured ETL pipelines for healthcare data
- Advanced SQL techniques (window functions, CTEs, subqueries)
- Combining clinical and environmental datasets for deeper insights

Overall, this analysis bridges **clinical knowledge and data analytics**, providing a foundation for:

- Risk stratification models
- Hospital resource planning
- Public health policy and early warning systems

Author:

Morteza Kouhestani || pharamacist ||, data scientist in healthcare

<https://github.com/Kouhestani-morteza>